The University of Hong Kong
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

COMP7103A DATA MINING
Date: 9th December 2024 Time: 6:30pm – 8:30pm

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is the candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

This is a closed-book exam.

Answer ALL 4 questions.

1. Short questions (30%)

   Give short answers to the following questions. Limit your answer of each part to not more than half a page.

   (a) (5%) Briefly describe how you would evaluate the similarity of two text documents. State the steps you would carry out.

   (b) (5%) In the context of decision tree classifier construction, explain the idea of pre-pruning and post-pruning. Also explain the objective of carrying out such tree pruning.

   (c) (5%) Explain "curse of dimensionality". Name two dimension reduction techniques.

   (d) (5%) Explain two techniques that are employed in SVM for handling data that is not linearly separable.

   (e) (5%) What is an ROC curve? Explain how one could compare two classifier models' performance based on their ROC curves.

   (f) (5%) Recall that in each iteration of the GSP algorithm, three steps are carried out, namely, (1) candidate generation, (2) candidate pruning, and (3) support counting. Given the following set of frequent 3-sequences

   $$\langle\{a,e\},\{b\}\rangle,\ \langle\{a,e\},\{c\}\rangle,\ \langle\{b\},\{a\},\{e\}\rangle,\ \langle\{e\},\{b\},\{a\}\rangle,\ \langle\{e\},\{b,d\}\rangle,$$

   determine the set of candidate 4-sequences generated by step (1). Show the resulting candidate 4-sequences after candidate pruning (step (2)).

2. (25%) Consider a dataset D of labeled objects shown in Table 1. In this table, each row shows an object id, two nominal attributes (a_1, a_2), one numerical attribute (a_3), and a class label.

Object	a_1	a_2	a_3	Class
X_1	True	High	6	$-$
X_2	True	High	8	$-$
X_3	True	Low	2	$-$
X_4	False	High	4	$+$
X_5	False	High	6	$+$
X_6	False	High	8	$-$
X_7	False	Low	4	$+$
X_8	False	Low	6	$+$

Table 1: Dataset D

(a) (5%) If D is split by a test on attribute a_1, calculate the expected Gini of the dataset after the split.

(b) (5%) If D is split by a binary test on a_3, find the best split point using classification error as the impurity measure.

(c) Consider applying boosting in classifier construction. Assume we use only attribute a_2 to build a decision stump (decision tree with only one test). Suppose in the first round of boosting the following examples are selected into a training sample.

$$S : X_4,\ X_6,\ X_3,\ X_7,\ X_1,\ X_5,\ X_7,\ X_2.$$

i. (5%) Draw the resulting decision stump trained with sample S.

ii. (5%) Show the classification results of applying the decision stump onto the 8 data objects of Table 1. Then, summarize the results in a confusion matrix.

iii. (5%) Explain how the sample weight for each object in Table 1 might change before the second round is conducted.

3. (20%) Table 2 shows the support and related information of some itemsets in a market basket dataset D. The set of all items is $\{a, b, c, d, e\}$.

Itemset	Support	Is Frequent?	Is Maximal Frequent?
$\{b\}$	55%	Yes	No
$\{c\}$	35%	Yes	Yes
$\{d\}$	50%	Yes	No
$\{a, b\}$	25%	No	No
$\{a, d\}$	25%	No	No
$\{a, e\}$	50%	Yes	Yes
$\{b, d\}$	30%	Yes	Yes
$\{b, e\}$	30%	Yes	Yes
$\{d, e\}$	30%	Yes	Yes
$\{b, d, e\}$	10%	No	No

Table 2: Support and information of some itemsets in D

(a) (5%) Calculate the Jaccard coefficient of $\{a, b\}$ and $\{b, d, e\}$.

(b) (5%) Calculate the confidence of the rule $\{b\} \rightarrow \{d\}$ with respect to D.

(c) (5%) If $\{a\}$ is not a closed itemset in D, what is the support of $\{a\}$ in D? Explain your answer.

(d) (5%) Consider applying the Apriori Algorithm to find frequent itemsets in D. Give the set of candidate 3-itemsets, C_3, generated by the algorithm.

4. (25%) Table 3 shows the coordinates of 5 data objects in a 2D space, as well as the distance matrix using Manhattan Distance as the distance measure.

Object	Coordinates	O_1	O_2	O_3	O_4	O_5
O_1	$(0, 24)$	0	24	27	12	42
O_2	$(0, 48)$	24	0	51	36	66
O_3	$(3, 0)$	27	51	0	33	63
O_4	$(12, 24)$	12	36	33	0	30
O_5	$(42, 24)$	42	66	63	30	0

Table 3: Distance matrix

(a) (8%) Figure 1 shows a partial dendrogram constructed using *agglomerative hierarchical clustering* with *complete link* as cluster similarity. Deduce the distance value A and object id B shown in the figure.

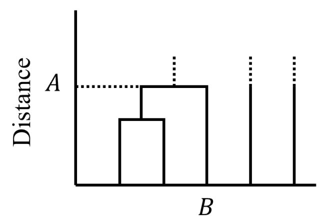

Figure 1: Partial dendrogram

(b) (8%) Consider clustering the objects in Table 3 using bisecting k-means. If O_1 and O_4 are selected as the initial centroids, compute the SSE of the clustering after the first iteration of the algorithm.

(c) (9%) Consider clustering the objects in Table 3 using DBSCAN with parameters $\epsilon = 32$ and $MinPts = 4$. Determine the ϵ-neighbors of each object. Then, identify all core points, border points, and noise points.

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